

## Problem Set 3

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1.

a) The null hypothesis is that there is no association between treatment and presence of rash after 24 hours for Center A or Center B. Under large sample theory,  $Q_{MH} \sim \chi^2_{(1)}$ .  $Q_{MH} = 5.07$  with 1 d.f. This is clearly significant. There is a strong association between treatment and presence of rash when adjusting for center. Active treatment had a significantly higher rash rate than placebo treatment.

The FREQ Procedure

Table 1 of treatment by rash  
Controlling for center=A

| treatment          |     | rash  |       |       |
|--------------------|-----|-------|-------|-------|
| Frequency          |     |       |       |       |
| Row                | Pct | Y     | N     | Total |
| -----+-----+-----+ |     |       |       |       |
| Active             |     | 52    | 26    | 78    |
|                    |     | 66.67 | 33.33 |       |
| -----+-----+-----+ |     |       |       |       |
| Placebo            |     | 35    | 40    | 75    |
|                    |     | 46.67 | 53.33 |       |
| -----+-----+-----+ |     |       |       |       |
| Total              |     | 87    | 66    | 153   |

Table 2 of treatment by rash  
Controlling for center=B

| treatment          |     | rash  |       |       |
|--------------------|-----|-------|-------|-------|
| Frequency          |     |       |       |       |
| Row                | Pct | Y     | N     | Total |
| -----+-----+-----+ |     |       |       |       |
| Active             |     | 45    | 34    | 79    |
|                    |     | 56.96 | 43.04 |       |
| -----+-----+-----+ |     |       |       |       |
| Placebo            |     | 40    | 38    | 78    |
|                    |     | 51.28 | 48.72 |       |

```

-----+-----+-----+
Total          85       72      157

```

Summary Statistics for treatment by rash  
Controlling for center

Cochran-Mantel-Haenszel Statistics (Based on Table Scores)

| Statistic | Alternative Hypothesis | DF | Value  | Prob   |
|-----------|------------------------|----|--------|--------|
| 1         | Nonzero Correlation    | 1  | 5.0677 | 0.0244 |
| 2         | Row Mean Scores Differ | 1  | 5.0677 | 0.0244 |
| 3         | General Association    | 1  | 5.0677 | 0.0244 |

Mantel-Fleiss Criterion      68.1236

b)    i.

Odds Ratio and Relative Risks for Center A

| Statistic                | Value  | 95% Confidence Limits |        |
|--------------------------|--------|-----------------------|--------|
| Odds Ratio               | 2.2857 | 1.1888                | 4.3949 |
| Relative Risk (Column 1) | 1.4286 | 1.0707                | 1.9061 |
| Relative Risk (Column 2) | 0.6250 | 0.4280                | 0.9126 |

The OR is 2.2857 with confidence limits (1.188,4.3949). Active treatment subjects had 2.2857 times the odds of developing a rash after 24 hours as placebo subjects. The confidence interval does not include 1 so there is a significant association between treatment and rash for center A.

ii.

Odds Ratio and Relative Risks for Center B

| Statistic                | Value  | 95% Confidence Limits |        |
|--------------------------|--------|-----------------------|--------|
| Odds Ratio               | 1.2574 | 0.6704                | 2.3582 |
| Relative Risk (Column 1) | 1.1108 | 0.8320                | 1.4830 |
| Relative Risk (Column 2) | 0.8834 | 0.6282                | 1.2422 |

The OR is 1.2574 with confidence limits (0.6704,2.3582). Active treatment subjects had 1.2574 times the odds of developing a rash after 24 hours as placebo subjects. The interval includes 1 so there is not a significant association between treatment and rash for center B.

iii.

Common Odds Ratio and Relative Risks

| Statistic  | Method          | Value  | 95% Confidence Limits |        |
|------------|-----------------|--------|-----------------------|--------|
| Odds Ratio | Mantel-Haenszel | 1.6760 | 1.0679                | 2.6304 |

|                          |                 |        |        |        |
|--------------------------|-----------------|--------|--------|--------|
|                          | Logit           | 1.6757 | 1.0650 | 2.6365 |
| Relative Risk (Column 1) | Mantel-Haenszel | 1.2601 | 1.0278 | 1.5449 |
|                          | Logit           | 1.2600 | 1.0274 | 1.5454 |
| Relative Risk (Column 2) | Mantel-Haenszel | 0.7500 | 0.5826 | 0.9657 |
|                          | Logit           | 0.7566 | 0.5873 | 0.9747 |

$\hat{\psi}_{MH} = 1.6760$  with confidence limits (1.0679, 2.6304) and  $\hat{\psi}_L = 1.6757$ . The confidence limits do not contain 1. For the combined centers, subjects with active treatment had 1.6760 times the odds of developing a rash as compared to placebo subjects.

c)

#### Breslow-Day Test for Homogeneity of Odds Ratios

```
-----
Chi-Square      1.6708
DF              1
Pr > ChiSq      0.1962
```

Output from Breslow-Day test is shown above.  $Q_{BD} = 1.6708$  with  $p = 0.1962$  so it does not contradict homogeneity of the odds ratio for these data. The null hypothesis is that the odds ratios for the 2 strata are equal. Under the null,  $Q_{BD}$  follows a chi-square distribution with 1 d.f.

2.

a) The assumptions are: a dichotomous response variable. In this case, the response variable is presence of rash after 24 hours. It is assumed that these data arise from a stratified simple random sample such that the presence or absence of rash is binomially distributed for each treatment  $\times$  center combination.

$$\text{logit}(\theta_{ij}) = \alpha + \beta_1 X_1 + \beta_2 X_2$$

where  $\theta_{ij}$  = probability that person of  $i$ th treatment with  $j$ th center has rash after 24 hours for  $i = 1, 2$  where 1=active treatment and 2 = placebo treatment and  $j = 1, 2$  where 1=Center A, 2=Center B

$\alpha$  = log odds of rash after 24 hours for placebo treatment and Center B

$\beta_1$  = increment in log odds for active treatment

$\beta_2$  = increment in log odds for Center B

$$X_1 = \begin{cases} 1 & \text{treatment} = \text{"Active"} \\ 0 & \text{treatment} = \text{"Placebo"} \end{cases}$$

$$X_2 = \begin{cases} 1 & \text{Center} = \text{A} \\ 0 & \text{Center} = \text{B} \end{cases}$$

b)

#### Odds Ratio Estimates

|       |          |
|-------|----------|
| Point | 95% Wald |
|-------|----------|

| Effect    |                   | Estimate | Confidence Limits |       |
|-----------|-------------------|----------|-------------------|-------|
| treatment | Active vs Placebo | 1.681    | 1.070             | 2.641 |
| center    | A vs B            | 1.115    | 0.709             | 1.752 |

The odds of developing a rash after 24 hours among those receiving active treatment were 1.681 times the odds of developing after 24 hours among those receiving placebo treatment after adjusting for center. The OR is 1.681 with confidence limits (1.070,2.641). As 1 is not contained in this interval, this indicates that there is a significant association between treatment and developing a rash after 24 hours.

c) Again, a Wald test will be used. Wald statistic follows  $N(0,1)$  under large sample theory. The null hypothesis is that the parameter for the interaction term is 0. From the output shown below, we see that the Wald Chi-Square is 1.6672 ( $p = 0.1966$ ). We fail to reject the null hypothesis. The interaction is not statistically significant. There is not evidence to suggest that the association between treatment and presence of rash after 24 hours depends on Center.

#### Analysis of Maximum Likelihood Estimates

| Parameter                 | DF | Estimate | Standard Error | Wald Chi-Square | Pr > ChiSq |
|---------------------------|----|----------|----------------|-----------------|------------|
| Intercept                 | 1  | 0.0513   | 0.2265         | 0.0513          | 0.8209     |
| treatment Active          | 1  | 0.2290   | 0.3209         | 0.5094          | 0.4754     |
| center A                  | 1  | -0.1848  | 0.3239         | 0.3257          | 0.5682     |
| treatment*center Active A | 1  | 0.5976   | 0.4628         | 1.6672          | 0.1966     |

d) Problem 1 suggested homogeneity of the odds ratios across centers. Problem 2 suggested the opposite. The non-significant p-value in 2 indicated that odds ratios were not homogeneous after all.

3.

```
data respiratory;
input status :$9. region :$7. response :$ count @@;
datalines;
favorable West disagree 26 favorable West neutral 14 favorable West agree 50
favorable Midwest disagree 15 favorable Midwest neutral 15 favorable Midwest agree 22
favorable East disagree 33 favorable East neutral 28 favorable East agree 37
mildly_u West disagree 48 mildly_u West neutral 35 mildly_u West agree 30
mildly_u Midwest disagree 16 mildly_u Midwest neutral 18 mildly_u Midwest agree 21
mildly_u East disagree 53 mildly_u East neutral 34 mildly_u East agree 28
mod_u West disagree 10 mod_u West neutral 18 mod_u West agree 15
mod_u Midwest disagree 7 mod_u Midwest neutral 17 mod_u Midwest agree 19
mod_u East disagree 20 mod_u East neutral 24 mod_u East agree 21
;
*proc print;
run;
```

```

*Combining mildly unfavorable and moderately unfavorable;
data respiratory2;
  set respiratory;
  if status ne '' and status in ('mildly_u','mod_u') then status_new = 'U';
  else status_new = 'F';
run;

proc freq data = respiratory2 order = data;
  weight count;
  tables region*status_new*response / cmh nocol nopercent scores=modridit;
run;

```

Table 1 of status\_new by response  
Controlling for region=West

| status_new               |     | response |         |       |       |
|--------------------------|-----|----------|---------|-------|-------|
| Frequency                |     |          |         |       |       |
| Row                      | Pct | disagree | neutral | agree | Total |
| -----+-----+-----+-----+ |     |          |         |       |       |
| F                        |     | 26       | 14      | 50    | 90    |
|                          |     | 28.89    | 15.56   | 55.56 |       |
| -----+-----+-----+-----+ |     |          |         |       |       |
| U                        |     | 58       | 53      | 45    | 156   |
|                          |     | 37.18    | 33.97   | 28.85 |       |
| -----+-----+-----+-----+ |     |          |         |       |       |
| Total                    |     | 84       | 67      | 95    | 246   |

Table 2 of status\_new by response  
Controlling for region=Midwest

| status_new               |     | response |         |       |       |
|--------------------------|-----|----------|---------|-------|-------|
| Frequency                |     |          |         |       |       |
| Row                      | Pct | disagree | neutral | agree | Total |
| -----+-----+-----+-----+ |     |          |         |       |       |
| F                        |     | 15       | 15      | 22    | 52    |
|                          |     | 28.85    | 28.85   | 42.31 |       |
| -----+-----+-----+-----+ |     |          |         |       |       |
| U                        |     | 23       | 35      | 40    | 98    |
|                          |     | 23.47    | 35.71   | 40.82 |       |
| -----+-----+-----+-----+ |     |          |         |       |       |
| Total                    |     | 38       | 50      | 62    | 150   |

Table 3 of status\_new by response  
Controlling for region=East

status\_new response

| Frequency                |          |         |       |       |
|--------------------------|----------|---------|-------|-------|
| Row Pct                  | disagree | neutral | agree | Total |
| -----+-----+-----+-----+ |          |         |       |       |
| F                        | 33       | 28      | 37    | 98    |
|                          | 33.67    | 28.57   | 37.76 |       |
| -----+-----+-----+-----+ |          |         |       |       |
| U                        | 73       | 58      | 49    | 180   |
|                          | 40.56    | 32.22   | 27.22 |       |
| -----+-----+-----+-----+ |          |         |       |       |
| Total                    | 106      | 86      | 86    | 278   |

### The FREQ Procedure

Summary Statistics for status\_new by response  
Controlling for region

Cochran-Mantel-Haenszel Statistics (Modified Ridit Scores)

| Statistic | Alternative Hypothesis | DF | Value   | Prob   |
|-----------|------------------------|----|---------|--------|
| -----     |                        |    |         |        |
| 1         | Nonzero Correlation    | 1  | 8.2685  | 0.0040 |
| 2         | Row Mean Scores Differ | 1  | 8.2580  | 0.0041 |
| 3         | General Association    | 2  | 15.0146 | 0.0005 |

a) Modified ridit scores were used since we have an ordinal response variable. The null hypothesis is that there is no association between respiratory status and agreement with new environmental policy when controlling for region. The alternative is that there are location shifts.  $Q_{SMH} = 8.2580$  ( $p = 0.0041$ ). The test statistic follows chi-square with 1 d.f. Since the p value is  $< 0.05$ , we reject the null and conclude that there are location shifts when controlling for region.

b) The result from a) was statistically significant. In all 3 regions, we saw a greater proportion of individuals with favorable health status agreeing with the new policy.

c)

```
data respiratory3;
  set respiratory;
  agree_b = response = 'agree'; /*Assigning boolean variable for agreement*/
run;
```

```
proc freq data = respiratory3 order = data;
  weight count;
  tables status*agree_b /cmh chisq measures trend nocol nopct;
  tables status*agree_b / cmh chisq scores=modridit nocol nopct;
  format agree_b agreef.;
run;
```

## The FREQ Procedure

Table of status by agree\_b

| status             | agree_b  |       |       |
|--------------------|----------|-------|-------|
| Frequency          |          |       |       |
| Row Pct            | Disagree | Agree | Total |
| -----+-----+-----+ |          |       |       |
| favorable          | 131      | 109   | 240   |
|                    | 54.58    | 45.42 |       |
| -----+-----+-----+ |          |       |       |
| mildly_u           | 204      | 79    | 283   |
|                    | 72.08    | 27.92 |       |
| -----+-----+-----+ |          |       |       |
| mod_u              | 96       | 55    | 151   |
|                    | 63.58    | 36.42 |       |
| -----+-----+-----+ |          |       |       |
| Total              | 431      | 243   | 674   |

Summary Statistics for status by agree\_b

Cochran-Mantel-Haenszel Statistics (Based on Table Scores)

| Statistic          | Alternative Hypothesis | DF | Value   | Prob   |
|--------------------|------------------------|----|---------|--------|
| -----+-----+-----+ |                        |    |         |        |
| 1                  | Nonzero Correlation    | 1  | 5.4834  | 0.0192 |
| 2                  | Row Mean Scores Differ | 2  | 17.2396 | 0.0002 |
| 3                  | General Association    | 2  | 17.2396 | 0.0002 |

Summary Statistics for status by agree\_b

Cochran-Mantel-Haenszel Statistics (Modified Ridit Scores)

| Statistic          | Alternative Hypothesis | DF | Value   | Prob   |
|--------------------|------------------------|----|---------|--------|
| -----+-----+-----+ |                        |    |         |        |
| 1                  | Nonzero Correlation    | 1  | 6.4651  | 0.0110 |
| 2                  | Row Mean Scores Differ | 2  | 17.2396 | 0.0002 |
| 3                  | General Association    | 2  | 17.2396 | 0.0002 |

The null hypothesis is there is no association between respiratory status and agreement with new environmental policy. The test statistic follows chi-square with 1 d.f. Integer scores produce a  $Q_{\text{CSMH}}$  of 5.4834 and modified ridit scores produce  $Q_{\text{CSMH}}$  of 6.4651. These scores are clearly significant. The proportion of individuals agreeing with environmental policy is correlated with respiratory health status with more individuals with favorable respiratory health status individuals more often in agreement with the new environmental policy.

d)

```
proc freq data = respiratory3 order = data;
  weight count;
  tables region*status*agree_b / cmh nocol nopercent;
  tables region*status*agree_b / cmh nocol nopercent scores=modridit;
  format agree_b agreef.;
run;
```

Table 1 of status by agree\_b  
Controlling for region=West

| status             | agree_b  |       |       |
|--------------------|----------|-------|-------|
| Frequency          |          |       |       |
| Row Pct            | Disagree | Agree | Total |
| -----+-----+-----+ |          |       |       |
| favorable          | 40       | 50    | 90    |
|                    | 44.44    | 55.56 |       |
| -----+-----+-----+ |          |       |       |
| mildly_u           | 83       | 30    | 113   |
|                    | 73.45    | 26.55 |       |
| -----+-----+-----+ |          |       |       |
| mod_u              | 28       | 15    | 43    |
|                    | 65.12    | 34.88 |       |
| -----+-----+-----+ |          |       |       |
| Total              | 151      | 95    | 246   |

Table 2 of status by agree\_b  
Controlling for region=Midwest

| status             | agree_b  |       |       |
|--------------------|----------|-------|-------|
| Frequency          |          |       |       |
| Row Pct            | Disagree | Agree | Total |
| -----+-----+-----+ |          |       |       |
| favorable          | 30       | 22    | 52    |
|                    | 57.69    | 42.31 |       |
| -----+-----+-----+ |          |       |       |
| mildly_u           | 34       | 21    | 55    |
|                    | 61.82    | 38.18 |       |
| -----+-----+-----+ |          |       |       |
| mod_u              | 24       | 19    | 43    |
|                    | 55.81    | 44.19 |       |
| -----+-----+-----+ |          |       |       |
| Total              | 88       | 62    | 150   |

3d

# The FREQ Procedure

Table 3 of status by agree\_b  
Controlling for region=East

| status             | agree_b  |       |       |
|--------------------|----------|-------|-------|
| Frequency          |          |       |       |
| Row Pct            | Disagree | Agree | Total |
| -----+-----+-----+ |          |       |       |
| favorable          | 61       | 37    | 98    |
|                    | 62.24    | 37.76 |       |
| -----+-----+-----+ |          |       |       |
| mildly_u           | 87       | 28    | 115   |
|                    | 75.65    | 24.35 |       |
| -----+-----+-----+ |          |       |       |
| mod_u              | 44       | 21    | 65    |
|                    | 67.69    | 32.31 |       |
| -----+-----+-----+ |          |       |       |
| Total              | 192      | 86    | 278   |

Summary Statistics for status by agree\_b  
Controlling for region  
Cochran-Mantel-Haenszel Statistics (Based on Table Scores)

| Statistic | Alternative Hypothesis | DF    | Value   | Prob   |
|-----------|------------------------|-------|---------|--------|
| -----     | -----                  | ----- | -----   | -----  |
| 1         | Nonzero Correlation    | 1     | 5.5431  | 0.0186 |
| 2         | Row Mean Scores Differ | 2     | 17.2565 | 0.0002 |
| 3         | General Association    | 2     | 17.2565 | 0.0002 |

Summary Statistics for status by agree\_b  
Controlling for region

Cochran-Mantel-Haenszel Statistics (Modified Ridit Scores)

| Statistic | Alternative Hypothesis | DF    | Value   | Prob   |
|-----------|------------------------|-------|---------|--------|
| -----     | -----                  | ----- | -----   | -----  |
| 1         | Nonzero Correlation    | 1     | 7.0249  | 0.0080 |
| 2         | Row Mean Scores Differ | 2     | 17.2706 | 0.0002 |
| 3         | General Association    | 2     | 17.2565 | 0.0002 |

The null hypothesis is there is no association between respiratory health status and agreement with new environmental policy across all region strata. The test statistic follows chi-square with 1 d.f. Integer scores produce a  $Q_{CSMH}$  of 5.5431 and modified ridit scores produce  $Q_{CSMH}$  of 7.0249. These scores are clearly significant. The proportion of individuals agreeing with new environmental

policy is correlated with respiratory health status when controlling for effect of region.

4. a)

$$\text{logit}(\theta_{ik}) = \alpha_k + \mathbf{x}_i' \boldsymbol{\beta}_k$$

where  $i = 1, 2, \dots, 6$  references the 6 populations determined by the levels of region and respiratory status and  $k$  indexes the logit.

$\theta_{i1}$  is the probability of agree

$\theta_{i2}$  is the probability of agree or neutral

$\alpha_1$  = log odds of agree versus neutral or disagree for east region and favorable respiratory status

$\alpha_2$  = log odds of agree or neutral versus disagree for east region and favorable respiratory status

$\beta_1$  = incremental effect due to west region

$\beta_2$  = incremental effect due to mildly unfavorable respiratory status

$\beta_3$  = incremental effect due to moderately unfavorable respiratory status

Assume the data arise from a stratified random sample. Individuals under study are like a random sample so that a multinomial distribution applies to the respective outcomes.  $\beta = \beta_k$  is a key assumption for this methodology. The effects in the model provide goodness of fit, in this case region and status. We are assuming no region and status interaction.

b)

```
proc format;
    invalue agreenf
        'agree'=1
        'neutral'=2
        'disagree'=3
    ;
run;

data resp_q4;
    set respiratory;
    where region ne 'Midwest'; /*ignoring Region='Midwest'*/
    *Derive numeric variable for agree to use for ordering;
    agreen=input(response,agreenf.);
run;

*Checking agreen derivation;
/* proc freq data = resp_q4; */
/*     tables response*agreen / list missing; */
/*     tables region; */
/* run; */

proc sort data = resp_q4 out=resp_q4sorted;
    by agreen;
run;

title '4b';
proc logistic data = resp_q4sorted order = data;
    weight count;
```

```

class region(ref='East') status(ref='favorable') / param=ref;
model response = region status / scale = none aggregate;
run;
title;

```

#### Analysis of Maximum Likelihood Estimates

| Parameter         | DF | Estimate | Standard Error | Wald Chi-Square | Pr > ChiSq |
|-------------------|----|----------|----------------|-----------------|------------|
| Intercept agree   | 1  | -0.4084  | 0.1605         | 6.4731          | 0.0110     |
| Intercept neutral | 1  | 0.8360   | 0.1640         | 25.9957         | <.0001     |
| region West       | 1  | 0.2974   | 0.1632         | 3.3184          | 0.0685     |
| status mildly_u   | 1  | -0.7779  | 0.1846         | 17.7643         | <.0001     |
| status mod_u      | 1  | -0.2200  | 0.2235         | 0.9684          | 0.3251     |

#### Score Test for the Proportional Odds Assumption

| Chi-Square | DF | Pr > ChiSq |
|------------|----|------------|
| 10.0288    | 3  | 0.0183     |

#### Deviance and Pearson Goodness-of-Fit Statistics

| Criterion | Value   | DF | Value/DF | Pr > ChiSq |
|-----------|---------|----|----------|------------|
| Deviance  | 13.7728 | 7  | 1.9675   | 0.0554     |
| Pearson   | 13.0947 | 7  | 1.8707   | 0.0698     |

Number of unique profiles: 6

The null hypothesis for GOF is that the model fit is adequate.  $Q_P$  is distributed as chi-square with  $(r-1)(s-1)-t$  where  $t$  is the number of explanatory variables. Here we have  $s = 6$  populations and  $r = 3$  categories and we have 3  $\beta$  so  $t = 3$ . Hence,  $(3-1)(6-1)-3 = 7$  df.  $Q_P = 13.0947$  ( $p = 0.0698$ ). With this p-value, we would reject the null hypothesis and assume that the model fit is not adequate.

For the proportional odds assumption, the null hypothesis is  $\beta = \beta_k$  for all  $k$ .  $Q_{RS}$  is 10.0288 with 3 df ( $p = 0.0183$ ). This is clearly significant so we would reject the null and conclude the assumption of proportional odds is not met for these data.

We can also do the score test with the interaction term in the model.

```

proc logistic order=data;
  weight count;
  class region(ref='East') status(ref='favorable') / param=ref;
  model response = region status region*status / selection = forward start = 2;
run;

```

#### Residual Chi-Square Test

|            |    |            |
|------------|----|------------|
| Chi-Square | DF | Pr > ChiSq |
| 1.4425     | 2  | 0.4861     |

The value of the test statistic is 1.4425 with  $p = 0.4861$ . This supports the adequacy of the main effects model. If the interaction term had been significant, we would include it in the model. To address departures from the proportional odds model, partial proportional odds model can be fit or if there is no proportionality, fit the model with different parameters for each of the cumulative logits.

c)

#### Odds Ratio Estimates

| Effect                       | Point<br>Estimate | 95% Wald<br>Confidence Limits |       |
|------------------------------|-------------------|-------------------------------|-------|
| region West vs East          | 1.346             | 0.978                         | 1.854 |
| status mildly_u vs favorable | 0.459             | 0.320                         | 0.660 |
| status mod_u vs favorable    | 0.803             | 0.518                         | 1.244 |

The odds ratio for West vs. East is 1.346 with confidence limits (0.978, 1.854).